

EDUCATION

2021 – 2026 **Ph.D., Physical Chemistry**, University of Oklahoma, Norman, OK.
 2016 – 2020 **B.S., Biochemistry**, University of Texas at Dallas, Richardson, TX.

WORK EXPERIENCE

July 2026 – Present **Postdoctoral Research Assistant**, Los Alamos National Laboratory (LANL)
 Advisor: Dr. Yu Zhang
 Summer 2023 **Graduate Student Internship**, LANL, Los Alamos, New Mexico.
 Advisor: Dr. Yu Zhang
 Fall 2021 – June 2026 **Graduate Research Assistant**, University of Oklahoma, Norman, OK.
 Advisor: Dr. Yihan Shao

FELLOWSHIPS & AWARDS

Spring 2025	Excellence in Graduate Research Award	University of Oklahoma
Spring 2023	Roger E. Frech Scholarship	University of Oklahoma
Spring 2023	Harwell Endowed Scholarship	University of Oklahoma
Summer 2021	Kenneth M. Nicholas Graduate Fellowship	University of Oklahoma
Fall 2016 – 2020	Academic Excellence Scholarship with Distinction	University of Texas at Dallas

RESEARCH EXPERIENCE (WITH VARIOUS COLLABORATORS)

2024 – Present	Descriptors for Chemical Reactivity in Surfaces and Metal Organic Frameworks Collaborators: Drs. Hyunho Noh (OU), Bin Wang (Tufts), Kasun Gunasooriya (OU), Yuezhi Mao (SDSU) ➤ Investigated scaling relations of adsorbates on metal surfaces using energy decomposition analysis, supporting older models with newer analysis techniques. ➤ Computationally supported hydrogen atom reactivity studies in Ce and Ti MOFs, relevant as catalytic descriptors and important for photoredox chemistry. ➤ Aided in the exploration of missing linker defects in Ti MOFs. EDA Scaling Relations MOF PCET
2023 - 2025	Investigation of Stereo Selectivity in Metal Catalyzed Furanosylation Reactions Collaborators: Drs. Indrajeet Sharma and Kenneth Nicholas (OU) ➤ Explored potential reaction mechanisms for metal catalyzed furanosylation reactions, which are important for synthesis of medicinal compounds. ➤ Rationalized the <i>cis</i>-selectivity in furanosylation product formation based on kinetic and thermodynamic relations. Organometallic Catalysis
2024 – 2025	Increased Photostability of Quantum Dots through Ligand Stabilization Collaborators: Dr. Yitong Dong (OU) ➤ Provided computational justification for the increased stabilization of perovskite quantum dots with small aromatic vs. larger aliphatic ligands. Quantum Dots Adsorption PWDFT GPW
2023 – Present	Excitonic Properties in Molecular Thin Films Collaborators: Drs. Madalina Furis (OU), Lloyd Bumm (OU), Sergei Tretiak and Yu Zhang (LANL) ➤ Investigated the excitonic coupling between dimers in organic semiconductor thin-films for photovoltaic materials. ➤ Proposed a rationale for exciton coupling strengths in different molecular crystal materials. Excitons Molecular Crystals
2021 – Present	Organometallic Catalysts Design and Mechanistic Insight Collaborators: Dr. Kenneth Nicholas (OU) ➤ Proposed potential catalysts and reaction mechanisms for the decomposition of the greenhouse gas nitrous oxide. ➤ Provided mechanistic insight into the diperoxo to monodioxo conversion in molybdenum porphyrin species, relevant to O₂ sequestration and oxygen atom transfer (OAT) reactions. Catalysis GHG OAT

† Indicates equal contribution.

PRE-PRINTS OR UNDER REVIEW

- (19) Pei, Z.; Wei, Z.; **Lander, C.**; Wickizer, C.; Ran, C.; Mao, Y.; Yang, Z.; Shao, Y. [Constructing Spin-Adiabatic States for the Modeling of Spin-Crossing Reactions. II. Analytic Gradient with Unrestricted Kohn-Sham Orbitals](#). **2025**.

PUBLICATIONS

- (18) Atteberry, M. L.; **Lander, C.**; Wickizer, C.; Wanasinghe, A.; Mi, C.; Akhmedov, N. G.; Xiang, S.; Gee, G. C.; Hadley, H.; Bryan, P. H.; Peters, J. W.; Furis, M.; Shao, Y.; Dong, Y. [Controlling the Surface Morphology of Strongly Confined CsPbBr₃ Perovskite Quantum Dots](#). *Nano Letters* **2026**, 26, 9418–9426.
- (17) **Lander, C. W.**; Herndon, J. W.; Zink, J.; Drena, A. S.; Pacheco, A. A.; Shao, Y.; Nicholas, K. M. [Mechanistic Pathways in the Thermal and Photochemical Conversion of \(TTP\)Mo\(\$\eta^2\$ -O₂\)₂ to \(TTP\)MoO₂ \(TTP = Tetratolylporphyrin\)](#). *Eur J Inorg Chem* **2026**, 29, e70264.
- (16) Wickizer, C.; **Lander, C.**; Pei, Z.; Yip, W. T.; Ran, C.; Shao, Y. [A Computational Modeling of ADLumin Chemiluminescence: Oxygenation and Dioxetanone Formation](#). *J Comput Chem* **2026**, 47, e70372.
- (15) Sengupta, S.; Pei, Z.; **Lander, C.**; Wickizer, C. R.; Homma, Y.; Afshari, H.; Mao, Y.; Huo, P.; Zhang, Y.; Tretiak, S.; Bumm, L. A.; Furis, M. I.; Shao, Y. [Towards a theoretical understanding of excitonic properties of phthalocyanine thin films. I. Low-temperature exciton absorption spectra](#). *Physical Chemistry Chemical Physics* **2026**, 28, 7021–7034.
- (14) Londoño-Salazar, J.; Ye, G.; **Lander, C. W.**; Kapfunde, T. A.; Yamamoto, N.; Akhmedov, N. G.; Zhang, Y.; Shao, Y.; Richter-Addo, G. B. [Nitric Oxide Insertion into a Metalloporphyrin–Carbon Bond](#). *JACS Au* **2026**, jacsau.5c01770.
- (13) Altınçekiç, N. G.; **Lander, C. W.**; Yu, J.; Roslend, A.; Shao, Y.; Noh, H. [Proton, Electron, and Hydrogen-Atom Transfer Thermodynamics of the Metal–Organic Framework, Ti-MIL-125, Are Intrinsically Correlated to the Structural Disorder](#). *J. Am. Chem. Soc.* **2025**, jacs.5c10498.
- (12) DiCenso, J. †; **Lander, C. †**; Pei, Z.; Sengupta, S.; Le, T.; Zhang, Y.; Gunasooriya, G. T. K. K.; Wang, B.; Mao, Y.; Shao, Y. [Toward an Understanding of Linear Scaling Relations through Energy Decomposition Analysis](#). *JACS Au* **2025**, 5, 3092–3103.
- (11) Mi, C.; Gee, G. C.; **Lander, C. W.**; Shin, D.; Atteberry, M. L.; Akhmedov, N. G.; Hidayatova, L.; DiCenso, J. D.; Yip, W. T.; Chen, B.; Shao, Y.; Dong, Y. [Towards non-blinking and photostable perovskite quantum dots](#). *Nat Commun* **2025**, 16, 204.
- (10) Singh, S. P.; Chaudhary, U.; **Lander, C. W.**; Daroczi, A.; Shao, Y.; Sharma, I. [Iron-Carbene-Mediated Catalytic Activation of Conventional Thioglycosides for Stereoselective 1,2- cis -Furanosylations](#). *ACS Catal.* **2025**, 15, 9886–9896.
- (9) Altınçekiç, N. G.; **Lander, C. W.**; Roslend, A.; Yu, J.; Shao, Y.; Noh, H. [Electrochemically Determined and Structurally Justified Thermochemistry of H atom Transfer on Ti-Oxo Nodes of the Colloidal Metal–Organic Framework Ti-MIL-125](#). *J. Am. Chem. Soc.* **2024**, 146, 33485–33498.
- (8) Pan, X. †; Snyder, R. †; Wang, J. N. †; **Lander, C. †**; Wickizer, C. †; Van, R.; Chesney, A.; Xue, Y.; Mao, Y.; Mei, Y.; Pu, J.; Shao, Y. [Training machine learning potentials for reactive systems: A Colab tutorial on basic models](#). *J Comput Chem* **2024**, 45, 638–647.
- (7) Ghosh, B.; Alber, A.; **Lander, C. W.**; Shao, Y.; Nicholas, K. M.; Sharma, I. [Catalytic Stereoselective 1,2- cis -Furanosylations Enabled by Enynal-Derived Copper Carbenes](#). *ACS Catal.* **2024**, 14, 1037–1049.
- (6) Ghosh, B.; Alber, A.; **Lander, C. W.**; Shao, Y.; Nicholas, K. M.; Sharma, I. [Catalytic Activation of Thioglycosides with Copper-Carbenes for Stereoselective 1,2- Cis -Furanosylations](#). *Org. Lett.* **2024**, 26, 9436–9441.
- (5) Ingram, Z. J.; **Lander, C. W.**; Oliver, M. C.; Altınçekiç, N. G.; Huang, L.; Shao, Y.; Noh, H. [Hydrogen Atom Binding Energy of Structurally Well-Defined Cerium Oxide Nodes at the Metal–Organic Framework–Liquid Interfaces](#). *J. Phys. Chem. C* **2024**, 128, 9556–9565.
- (4) Rusmore, T. A.; **Lander, C.**; Nicholas, K. M. [Reaction Pathway for the Aerobic Oxidation of Phosphines Catalyzed by Oxomolybdenum Salen Complexes](#). *Eur J Inorg Chem* **2024**, 27, e202300506.
- (3) **Lander, C. †**; Satalkar, V. †; Yang, J. †; Pan, X.; Pei, Z.; Chatterji, A.; Liu, C.; Nicholas, K. M.; Cichewicz, R. H.; Yang, Z.; Shao, Y. [Visualization of electron density changes along chemical reaction pathways](#). *Molecular Physics* **2023**, 121, e2113566.
- (2) Jana, A. K.; **Lander, C. W.**; Chesney, A. D.; Hansmann, U. H. E. [Effect of an Amyloidogenic SARS-COV-2 Protein Fragment on \$\alpha\$ -Synuclein Monomers and Fibrils](#). *J. Phys. Chem. B* **2022**, 126, 3648–3658.
- (1) Nicholas, K. M.; **Lander, C.**; Shao, Y. [Computational Evaluation of Potential Molecular Catalysts for Nitrous Oxide Decomposition](#). *Inorg. Chem.* **2022**, 61, 14591–14605.

PRESENTATIONS

Oral	ACS Southwest Regional Meeting (2023)
Oral	ACS San Diego (2025)
Poster	ACS Southwest Regional Meeting (2024)
Poster	ACS New Orleans (2024)

TECHNICAL SKILLS

Software	Q-Chem, Gaussian, CP2K, VASP, PySCF, LAMMPS, ASE
Languages	Python, C/C++, FORTRAN

TEACHING

2025	Physical Chemistry I Lecture
2025	Physical Chemistry I Lab Teaching Assistant
2021 – 2022	General Chemistry II Lab and Recitation Teaching Assistant

MEMBERSHIP AND OUTREACH

2022 – 2024	<i>Graduate Student Senate</i> , University of Oklahoma
2022 – 2024	<i>Chemistry and Biochemistry Researchers</i> , University of Oklahoma Secretary (2022-2023) and Undergraduate Involvement Committee Chair (2023-2024)
2022 – 2024	<i>Graduate Student Recruitment Committee</i> , University of Oklahoma